

| Model          | Crop   | Accuracy |        | Precision |        | Recall |        | F1 Score |        |
|----------------|--------|----------|--------|-----------|--------|--------|--------|----------|--------|
|                |        | FP32     | INT8   | FP32      | INT8   | FP32   | INT8   | FP32     | INT8   |
| EfficientNetB2 | Apple  | 0.8615   | 0.8462 | 0.8658    | 0.8531 | 0.8615 | 0.8462 | 0.8571   | 0.8401 |
| EfficientNetB2 | Potato | 0.7667   | 0.7667 | 0.7727    | 0.7727 | 0.7667 | 0.7667 | 0.7615   | 0.7615 |
| VGG19          | Apple  | 0.8769   | 0.8769 | 0.8846    | 0.8846 | 0.8769 | 0.8769 | 0.8707   | 0.8707 |
| VGG19          | Potato | 0.7444   | 0.7444 | 0.7436    | 0.7436 | 0.7444 | 0.7444 | 0.7432   | 0.7432 |
| ResNet101      | Apple  | 0.4923   | 0.4923 | 0.4621    | 0.4621 | 0.4923 | 0.4923 | 0.4060   | 0.4060 |
| ResNet101      | Potato | 0.5889   | 0.5889 | 0.7637    | 0.7637 | 0.5889 | 0.5889 | 0.4675   | 0.4675 |
| InceptionV3    | Apple  | 0.7846   | 0.7538 | 0.7964    | 0.7606 | 0.7846 | 0.7538 | 0.7820   | 0.7504 |
| InceptionV3    | Potato | 0.7444   | 0.7333 | 0.7436    | 0.7325 | 0.7444 | 0.7333 | 0.7432   | 0.7325 |
| MobileNetV2    | Apple  | 0.6308   | 0.6308 | 0.6392    | 0.6392 | 0.6308 | 0.6308 | 0.6302   | 0.6302 |
| MobileNetV2    | Potato | 0.6000   | 0.6000 | 0.5945    | 0.5945 | 0.6000 | 0.6000 | 0.5861   | 0.5861 |
| DenseNet121    | Apple  | 0.7538   | 0.7692 | 0.7470    | 0.7823 | 0.7538 | 0.7692 | 0.7343   | 0.7483 |
| DenseNet121    | Potato | 0.7444   | 0.7333 | 0.7590    | 0.7496 | 0.7444 | 0.7333 | 0.7344   | 0.7214 |

Table 1: FP32 vs INT8 Performance Comparison for Apple and Potato Crop Disease Classification